Macro Microcontroller 2 Datasheet

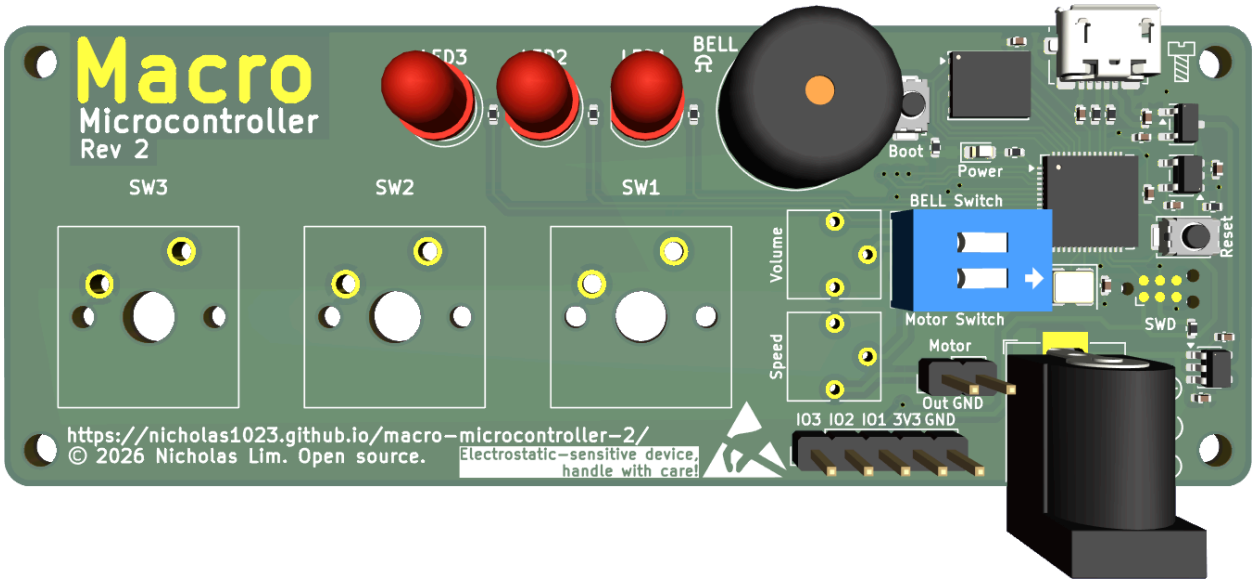


Fig 1: Top View of Macro Microcontroller 2

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Overview

Macro Microcontroller 2 is an RP2040-based microcontroller development board with an integrated macropad which provides GPIO controls with keyboard keys on a single printed circuit board, plus a custom BASIC interpreter and disk operating system.

Specifications

Microcontroller Unit: RP2040
Power: 5V input via data port or barrel jack, 3.3V input via GPIO pin headers, maximum logical voltage is 3.3V.
Dimensions: 96.52 × 35.56 mm
Flash Memory: 16 MB
Mounting Hole Dimensions: Ø 2.5mm

Pinout

Macro Microcontroller 2 provides a few IO pinouts for connecting peripherals.

- DC Motor (MOTOR)
- 2.54mm GPIO pin headers (IO3, IO2, IO1, 3.3V, GND)

Macro Microcontroller 2 Pin #	Corresponding RP2040 Pin #
IO1	GPIO3
IO2	GPIO4
IO3	GPIO5

Onboard Input/Output Devices

- 3 Key switches (SW1, SW2, SW3)
- 3 LEDs (LED1, LED2, LED3)
- 1 Buzzer (BELL)
- 1 DC motor port (MOTOR)
- 3 General purpose IO pins (IO1, IO2, IO3)

Enclosure

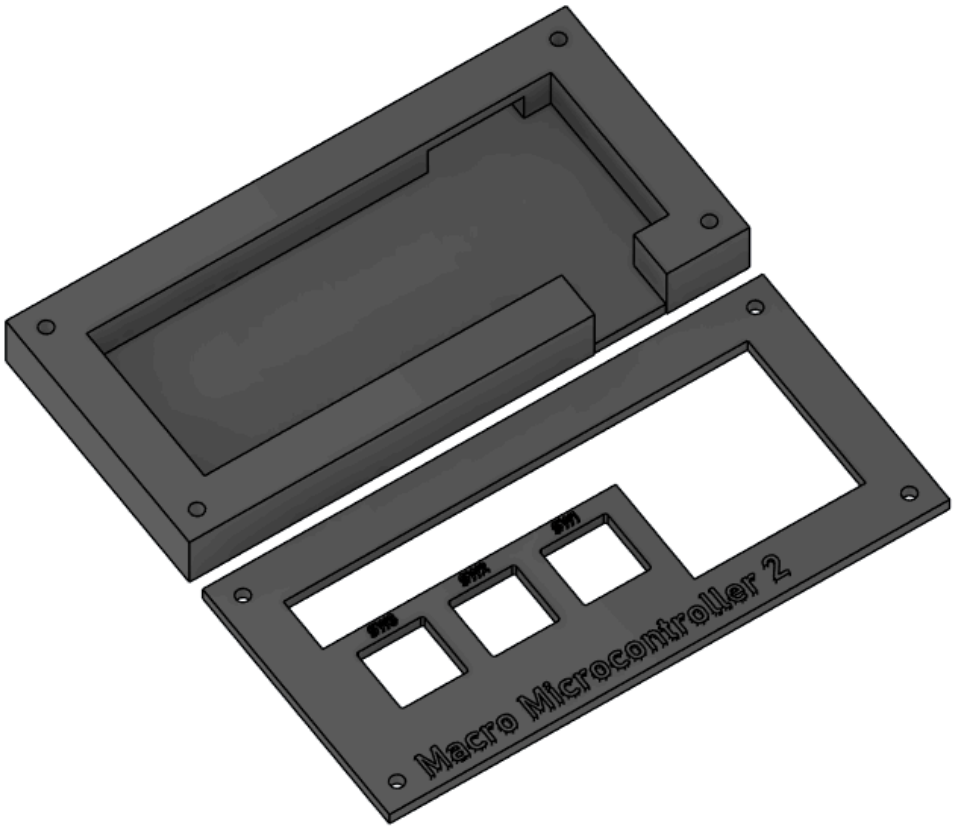


Fig 2: 3D Model of Macro Microcontroller 2 Enclosure

This enclosure is 3D-printable and is available for download from https://github.com/Nicholas1023/macro-microcontroller-2/releases/download/v0.0.1/macro-microcontroller-2-enclosure.stl

Schematics

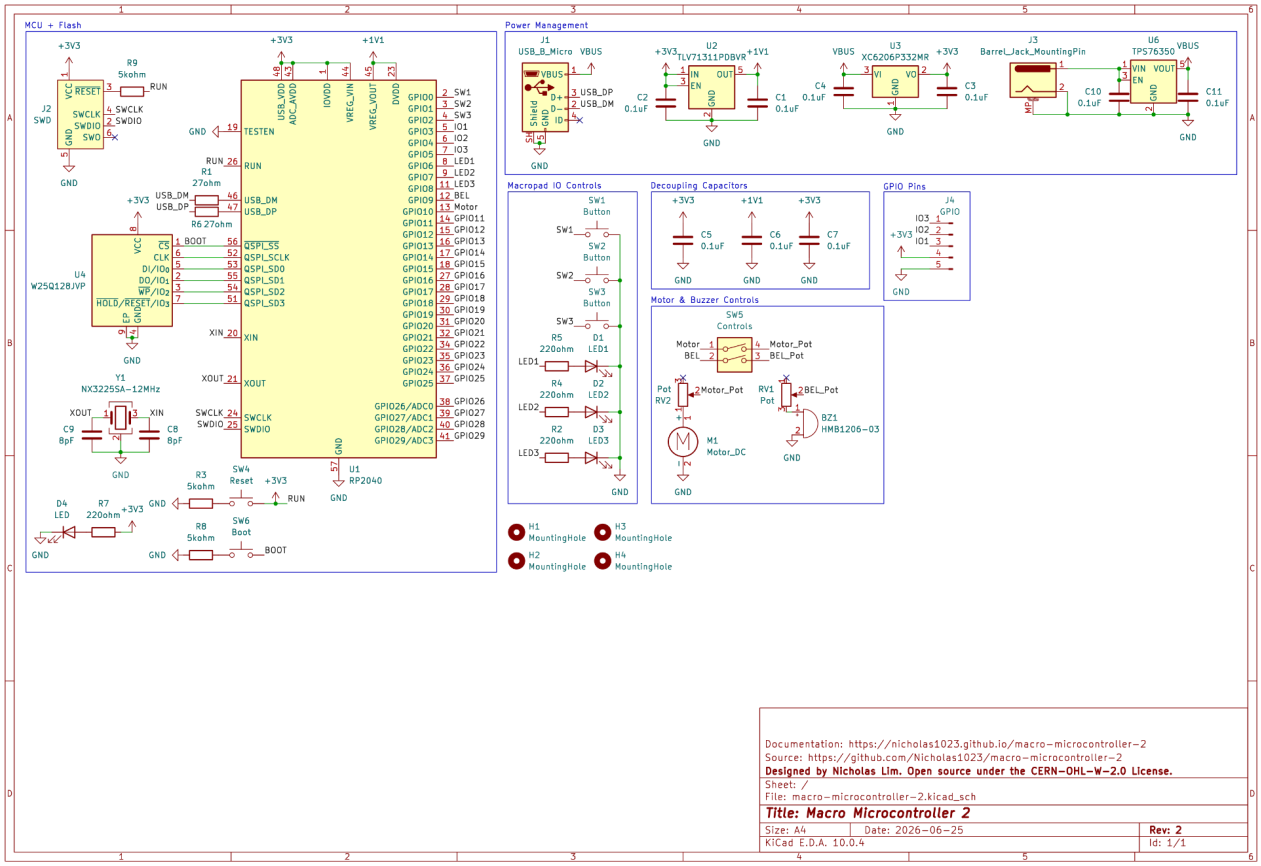


Fig 3: Schematics of Macro Microcontroller 2

PCB Routing and Layout

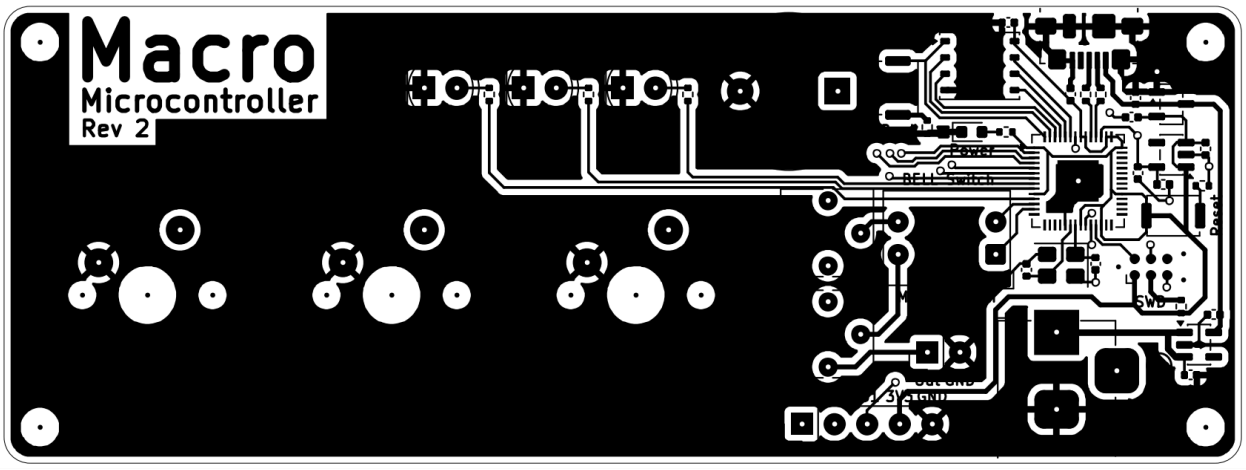


Fig 4: Copper traces and Silkscreen of Macro Microcontroller 2

Block Diagram

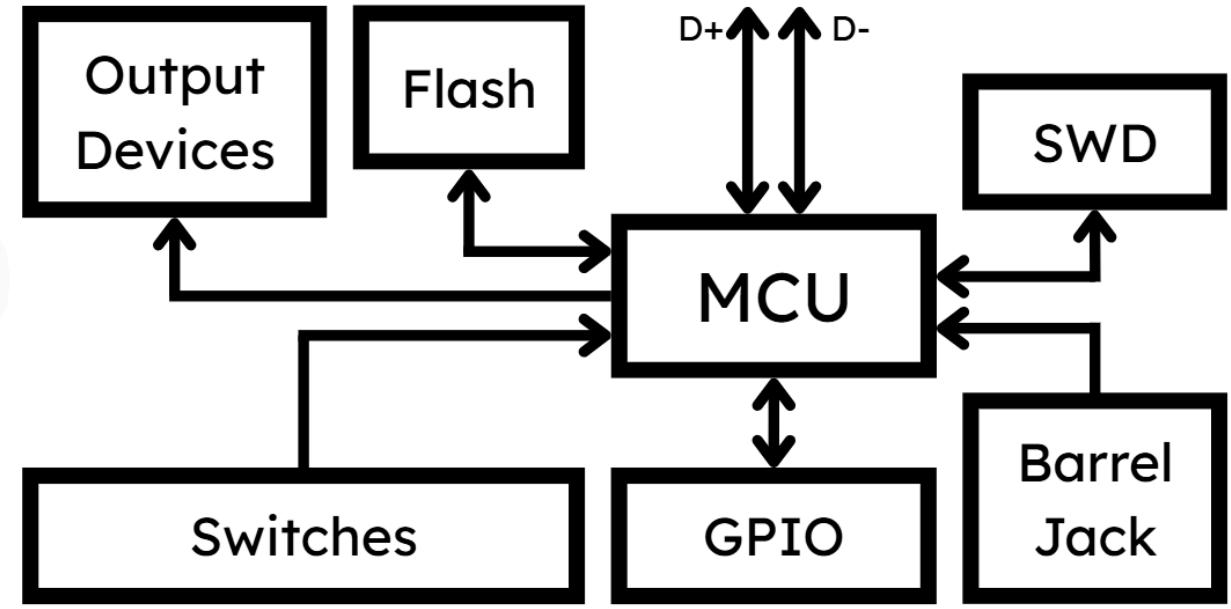


Fig 5: Block Diagram of Macro Microcontroller 2

Bill of Materials

Qty	Reference Designator	Part Description
1	BZ1	Buzzer
1	J3	Barrel Jack
1	SW5	DIP Switch
1	M1, J4	Pin Header
1	J1	Data Port
11	C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11	Capacitors
1	U1	RP2040
3	U2, U3, U6	Low Dropout Regulators
9	R1, R2, R3, R4, R5, R6, R7, R8, R9	Resistors
1	U4	Flash Memory
4	D1, D2, D3, D4	LED
1	Y1	Crystal Oscillator
2	RV1, RV2	Potentiometer
3	SW1, SW2, SW3	Keyboard Switch
2	SW4, SW6	Push Button

Macro Microcontroller DOS

Macro Microcontroller Disk Operating System (DOS) is a simple command-line operating system providing basic programs for Macro Microcontroller 2.

Installation

1. Download macro-microcontroller-dos-0.0.3.uf2. This release includes Macro Microcontroller BIOS, A basic BIOS-like firmware, which performs POST routines, reports information on the device and boot the DOS.

   SHA-256 Checksum:
   `e0ba748dbac0241309680e81cc556e139e9c126062bc54f37d4ad093bcaae639`
2. Plug in Macro Microcontroller 2 while holding the Boot button. A storage drive will appear on your computer.
3. Move macro-microcontroller-dos-0.0.3.uf2 into the storage drive. Your development board is now running Macro Microcontroller BIOS and DOS.
4. Install a serial terminal application on your computer and connect it to the board. If successful, you should see the following:

```
 _ _
| \/ | Macro Microcontroller BIOS Version 0.0.2.
|_\/_| Copyright (C) 2025-2026 Nicholas Lim.

Device: Macro Microcontroller 2 (RP2040 @ 125MHz)
Serial Number: ****************
RAM Free: 254K OK
Serial Status: Connected

Press SW1 to enter Macro Microcontroller BASIC.
Press SW2 to enter BOOTSEL mode.

Booting system...

Macro Microcontroller Disk Operating System Version 0.0.3.
Copyright (C) 2026 Nicholas Lim.
Drive A>
```

Commands and Programs

- `HELP`: Launches the help guide.
- `CALCULATE`: Performs basic operations on 2 numbers. Usage: `CALCULATE [Number] [+ - * / ^ root] [Number]`
- `ECHO`: Echoes text.
- `PROGLIST`: Show program listing.
- `SYSTEM`: Starts another Macro Microcontroller DOS console.
- `EXIT`: Stops all processes for power off.
- `OUT`: Turns on a specified GPIO for 1 second. Usage: `OUT [LED# IO# BELL MOTOR]`
- `VERSION`: Shows the release version.
- `BOOT`: Reboots the system. Usage: `BOOT -m [BASIC BOOTSEL]*`
- `DEVINFO`: Displays device and memory information.
- `UPTIME`: Shows device uptime.

Predefined Values for Parameters

- `IO*`: Set GPIO pins 1, 2 or 3 to high.
- `LED*`: Set LEDs 1, 2 or 3 to high.
- `BELL`: Sounds the onboard buzzer.
- `MOTOR`: Sets motor pins to high.
- `BASIC`: Reboot in BASIC interpreter mode.
- `BOOTSEL`: Reboot in BOOTSEL mode.

Macro Microcontroller BASIC

Macro Microcontroller BASIC is a custom BASIC interpreter written in C with low-level hardware communication, providing a programming environment via a serial terminal.

Installation

1. Download macro-microcontroller-basic-0.0.3.uf2.

   SHA-256 Checksum:
   `96d4aa58d61d796067c1668cc37896fa968a55dceea188e50f26e067d116e99f`
2. Plug in Macro Microcontroller 2 while holding the Boot button. A storage drive will appear on your computer.
3. Move macro-microcontroller-basic-0.0.3.uf2 into the storage drive. Your development board is now running Macro Microcontroller BASIC.
4. Install a serial terminal application on your computer and connect it to the board. If successful, you should see the following:
   ```
   Macro Microcontroller BASIC Version 0.0.3.
   Copyright (C) 2025-2026 Nicholas Lim.
   ****** bytes of RAM free.
   Ready.
   10
   ```
5. If not, try sending `PRINT "It works!"`. If "It works!" is sent back to the serial terminal, you have successfully installed Macro Microcontroller BASIC.

Statements

- `PRINT`: Outputs text to the serial terminal.
- `LET`: Assigns a value (integer, float or string) to a variable.
- `INPUT`: Assigns a value (integer, float or string) to a variable via user input.
- `GPIO`: Turns on a specified GPIO pin for 1 second.
- `LSVAR`: Outputs variable name and their value to the serial terminal.
- `END`: Clears variable and reset line numbers.
- `EXIT`: Puts the device into sleep/power off mode.
- `KEY`: Pass GPIO controls to the macropad.
- `REM`: Add comments to your program.

Predefined Values

- `IO*`: Set GPIO pins 1, 2 or 3 to high.
- `LED*`: Set LEDs 1, 2 or 3 to high.
- `BELL`: Sounds the onboard buzzer.
- `MOTOR`: Sets motor pins to high.
- `RND`: Generates a random floating-point integer less than 1.

Precautions

- Do not apply logical voltages exceeding 3.3V to any GPIO or inputs. This may cause onboard components to be damaged.
- Use only one power source at a time, either via the data port, barrel jack or 3V3 and GND pins to power the board.
- Only the data port and barrel jack accepts a maximum voltage of 5V. Onboard Low Dropout (LDO) Regulators provide regulated 3.3V when powered by either inputs only. The 3V3 pin provides a separate power connection and bypasses these input regulators.
- This is an electrostatic-sensitive device. Handle the board only by the sides and avoid touching components and pins.
- Do not handle with wet hands.

For more information on Macro Microcontroller 2, visit:
https://github.com/Nicholas1023/macro-microcontroller-2 (Repository)
https://nicholas1023.github.io/macro-microcontroller-2 (Documentation site)

Version History

Version 3, Revision B: Update description, reorder headers and simplify bill of materials
Version 3, Revision A: Add disclaimer and modify formatting
Version 2, Revision F: Rephrase precaution section
Version 2, Revision E: Update command listing for Macro Microcontroller DOS
Version 2, Revision D: Change SHA-256 checksum for Macro Microcontroller BIOS
Version 2, Revision C: Change overview and licensing section
Version 2, Revision B: Update schematics
Version 2, Revision A: Change overview and specification section and add Macro Microcontroller DOS section
Version 1, Revision B: Change in page number and URLs
Version 1, Revision A: Initial revision

Legal

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